

A Power Model for D-MIMO Systems: Comparing the Energy Consumption of D-MIMO and Co-located MIMO

Thomas Feys, François Rottenberg
KU Leuven, Dept. ESAT-WaveCore, Dramco, B-9000 Ghent, Belgium

Abstract—TODO
Index Terms—

I. INTRODUCTION

TODO small paragraph on energy consumption of 5G 6G networks

Recent works have focused on characterizing this energy consumption of wireless networks, and in particular of multiple input multiple output (MIMO) systems. In [1], the authors propose a detailed power consumption model for collocated MIMO systems, which is based on the power consumption of the different hardware components in the system for frequency range 3 (FR3) which is envisioned for 6G systems. Another prevelant enabler for 6G systems is distributed MIMO (D-MIMO)/cell-free MIMO. However, works that characterize the power consumption of D-MIMO systems is limited. **TODO add som refs and work here** Moreover, it is unclear whether the power consumption of D-MIMO systems is on par, lower or higher than the one of collocated MIMO systems. Especially, considering the additional network consumption that occurs from having multiple access point (AP), fronthaul links, synchronization, etc. In this work we introduce an extension of the power model introduced in [1] for the D-MIMO scenario. We incorporate **TODO mention all extensions. TODO mention it is opensource an where it can be found.**

II. D-MIMO SYSTEM MODEL

We consider a time division duplexing (TDD) D-MIMO network in which L APs, each with M antennas, jointly serve K single-antenna user equipments (UEs) over Q data orthogonal frequency division multiplexing (OFDM) subcarriers, resulting in LM total antennas. All L APs are connected to a central processing unit (CPU) by fronthaul links. The indices $l = 1, \dots, L$, $m = 1, \dots, M$, $k = 1, \dots, K$ and $q = 1, \dots, Q$ denote the number of APs, antennas per AP, UEs and subcarriers respectively. Co-located massive MIMO (mMIMO) is the special case $L = 1$ with $M_{\text{ant}} = LM$ antennas on a single site at the centre of the coverage area. Co-located and distributed MIMO consider the same number of total antennas and radiate the same total power, they only differ in where the antennas sit.

We consider the same frame structure as in [1]. Namely, the total frame length is T_f which is split into a downlink (DL) and

uplink (UL) phase of length $\tau_{\text{DL}}T_f$ and $\tau_{\text{UL}}T_f$ respectively, with $\tau_{\text{DL}}, \tau_{\text{UL}} \in [0, 1]$. We consider N_{DL} available OFDM symbols in DL and N_{UL} in UL, of which $N_{\text{a,DL}}$ and $N_{\text{a,UL}}$ are activated and carrying data. The activated fraction is the average physical resource load

$$\bar{x}_i = \frac{N_{\text{a},i}}{N_i} \in [0, 1] \quad i \in \{\text{DL}, \text{UL}\}, \quad (1)$$

so $\bar{x}_i = 1$ is a fully loaded direction and $\bar{x}_i = 0$ one that carries reference signals only. In general, as defined in [1], the load also counts how many UEs are served and how many subcarriers each of them is allocated. We assume throughout that every activated symbol serves all K UEs on all Q subcarriers, so the load reduces to the ratio of activated to available symbols. The UL and DL frames are further subdivided into three parts as depicted in **TODO add fig and ref it**. The first part is for reference signaling and has length

$$\tau_i \tau_{\text{sig},i} \quad i \in \{\text{DL}, \text{UL}\}, \quad (2)$$

where $\tau_{\text{sig},i} \in [0, 1]$ is the fraction of the frame used for signaling. The second part is for data transmission and has length

$$\tau_i(1 - \tau_{\text{sig},i})\bar{x}_i \quad i \in \{\text{DL}, \text{UL}\}. \quad (3)$$

The third part represents a micro-sleep or discontinuous transmission (DTX) mode [2] and has length

$$\tau_i(1 - \tau_{\text{sig},i})(1 - \bar{x}_i) \quad i \in \{\text{DL}, \text{UL}\}. \quad (4)$$

A. Signal Model

Let $\mathbf{h}_{kl}[q] \in \mathbb{C}^M$ be the channel from AP l to UE k . Stacking the APs gives the collective channel $\mathbf{h}_k[q] = [\mathbf{h}_{k1}[q]^\top \dots \mathbf{h}_{kL}[q]^\top]^\top \in \mathbb{C}^{LM}$ and combining the users gives $\mathbf{H}[q] = [\mathbf{h}_1[q] \dots \mathbf{h}_K[q]] \in \mathbb{C}^{LM \times K}$.

1) *Downlink*: AP l transmits $\mathbf{x}_l[q] = \sum_k \mathbf{w}_{kl}[q]s_k[q]$, with $\mathbf{w}_{kl}[q] \in \mathbb{C}^M$ its precoding vector for UE k and $s_k[q]$ the unit-power data symbols, mutually independent across UEs. Let $\mathbf{w}_k[q] \in \mathbb{C}^{LM}$ be the collection of the L precoders of the APs $\mathbf{w}_{kl}[q]$, the signal received by UE k is

$$y_k[q] = \mathbf{h}_k[q]^H \mathbf{w}_k[q]s_k[q] + \sum_{i \neq k} \mathbf{h}_k[q]^H \mathbf{w}_i[q]s_i[q] + n_k[q] \quad (5)$$

with $n_k[q] \sim \mathcal{CN}(0, \sigma^2/Q)$. Treating the multi-user interference as noise,

$$\text{SINR}_k^{\text{DL}}[q] = \frac{|\mathbf{h}_k[q]^H \mathbf{w}_k[q]|^2}{\sum_{i \neq k} |\mathbf{h}_k[q]^H \mathbf{w}_i[q]|^2 + \sigma^2/Q}, \quad (6)$$

where σ^2 is the receiver noise power.

2) *Uplink*: UE k transmits with power $p_k[q] = p_k/Q$ subject to its own budget $p_k \leq p_{\max}$, and AP l observes

$$\mathbf{y}_l[q] = \sum_k \sqrt{p_k[q]} \mathbf{h}_{kl}[q] s_k[q] + \mathbf{n}_l[q] \quad (7)$$

with $\mathbf{n}_l[q] \sim \mathcal{CN}(\mathbf{0}, (\sigma^2/Q)\mathbf{I}_M)$. Applying a receive combiner $\mathbf{v}_k[q] \in \mathbb{C}^{LM}$ gives

$$\hat{s}_k[q] = \mathbf{v}_k[q]^H \mathbf{y}_l[q] = \sum_{l=1}^L \mathbf{v}_{kl}[q]^H \mathbf{y}_l[q]. \quad (8)$$

The resulting signal-to-interference-plus-noise ratio (SINR) is

$$\text{SINR}_k^{\text{UL}}[q] = \frac{p_k |\mathbf{v}_k[q]^H \mathbf{h}_k[q]|^2}{\sum_{i \neq k} p_i |\mathbf{v}_k[q]^H \mathbf{h}_i[q]|^2 + \sigma^2 \|\mathbf{v}_k[q]\|^2}. \quad (9)$$

B. Channel and Noise Model

Channels are drawn from the 3GPP TR 38.901 urban microcell model [3] through the Sionna implementation [4]. Each AP is a 38.901 base station carrying an M -element half-wavelength uniform linear array (ULA) of omnidirectional elements at height h_{AP} , and each UE is a single-antenna outdoor terminal at height h_{UE} . UEs are sampled uniformly over a square coverage area of size $D \times D$ m², as are the APs. In the co-located case a single base station (BS) is placed at the centre. The receiver noise power follows from the thermal noise density, σ^2 [dBm] = $-174 + 10 \log_{10}(B/\text{Hz}) + F$, with B the bandwidth and F the noise figure in dB.

The large-scale fading coefficient β_{kl} , used for power control, is estimated as

$$\beta_{kl} = \frac{1}{MQ} \sum_{q=1}^Q \|\mathbf{h}_{kl}[q]\|^2. \quad (10)$$

C. Centralized and Distributed Operation

Each AP has its own power constraint. Since the symbols are independent and unit-power, the power radiated by AP l over the OFDM block is

$$P_{T,l} = \sum_{q=1}^Q \sum_{k=1}^K \|\mathbf{w}_{kl}[q]\|^2 \leq \rho_{\max}. \quad (11)$$

How this power constraint is met is the difference between centralized and distributed operation, this also depends on the functional splits of Section III-A2.

1) *Centralized operation (splits S1, S2)*: The CPU estimates the full channel matrix $\mathbf{H}[q]$ and designs all precoders jointly. In the downlink it applies zero forcing (ZF),

$$\bar{\mathbf{W}}[q] = \mathbf{H}[q](\mathbf{H}[q]^H \mathbf{H}[q])^{-1} \in \mathbb{C}^{LM \times K}, \quad (12)$$

which nulls all inter-user interference and requires $K \leq LM$. A single power coefficient ρ_k is attached to each UE, since the precoding direction already fixes how that power spreads over the LM antennas. Fractional power control weights the UEs by their aggregate gain $\beta_k = \sum_l \beta_{kl}$,

$$\rho_k = P_{\text{tot}} \frac{\beta_k^v}{\sum_i \beta_i^v}, \quad P_{\text{tot}} = L\rho_{\max}, \quad (13)$$

with the exponent v trading throughput against fairness¹. This power normalization is applied as

$$\mathbf{w}_k[q] = \gamma \sqrt{\rho_k} \frac{\bar{\mathbf{w}}_k[q]}{\sqrt{\sum_{q'} \|\bar{\mathbf{w}}_k[q']\|^2}}, \quad \gamma = \sqrt{\frac{\rho_{\max}}{\max_l \tilde{p}_l}}, \quad (14)$$

where $\tilde{p}_l$ is the power AP l would radiate before that rescaling. Only the busiest AP then reaches its budget. Since $\sum_k \rho_k = L\rho_{\max}$ the nominal per-AP powers average to $\rho_{\max}$, so $\gamma \leq 1$, the constraint is always met. The uplink counterpart forwards the raw observations to the CPU, which combines over all LM antennas with $\mathbf{V}[q] = \mathbf{H}[q](\mathbf{H}[q]^H \mathbf{H}[q])^{-1}$.

2) *Distributed operation (split S3)*: Each AP designs its own precoder/combiner from its local channel state information (CSI). Writing $\mathbf{E}_l[q] = [\mathbf{h}_{1l}[q] \cdots \mathbf{h}_{Kl}[q]] \in \mathbb{C}^{M \times K}$ for the local channel matrix and $\mathbf{P} = \text{diag}(p_1, \dots, p_K)$,

$$\mathbf{W}_l[q] = (\mathbf{E}_l[q]\mathbf{E}_l[q]^H + \lambda \mathbf{I}_M)^{-1} \mathbf{E}_l[q], \quad (15)$$

$$\mathbf{V}_l[q] = (\mathbf{E}_l[q]\mathbf{P}\mathbf{E}_l[q]^H + \lambda \mathbf{I}_M)^{-1} \mathbf{E}_l[q], \quad (16)$$

are the local regularized zero forcing (RZF) precoder/combiner which correspond to the local minimum mean squared error (MMSE) precoder/combiner, with $\lambda = \sigma^2$. Each AP also allocates its own power budget among the UEs from local statistics,

$$\rho_{kl} = \rho_{\max} \frac{\beta_{kl}^v}{\sum_i \beta_{il}^v}, \quad (17)$$

so $\sum_k \rho_{kl} = \rho_{\max}$ by construction and every AP meets (11) with equality, $P_{T,l} = \rho_{\max}$, without any CPU-side rescaling. In the uplink the CPU fuses the L local estimates by summing them, $\hat{s}_k[q] = \sum_l \mathbf{v}_{kl}[q]^H \mathbf{y}_l[q]$.

In the uplink fractional power control is the same for both centralized and distributed operation at $p_k = \rho_{\max} \beta_k^{v_{\text{ul}}} / \max_i \beta_i^{v_{\text{ul}}}$. This is normalized by a maximum rather than a sum because each UE has its own independent power budget.

¹A positive v follows the channel and serves the UEs with the largest aggregate gain first, which maximizes the sum rate but starves those at the edge of the coverage area. A negative v compensates the weak UEs: at $v = -1$ the product $\rho_k \beta_k$ is the same for every UE, so all of them are received at approximately equal signal power, which is the fair extreme. Values in $[-1, 1]$ therefore interpolate between a throughput-optimal and a max-min-fair allocation.

D. Delivered Rates

The per-UE spectral efficiencies average over the subcarriers and over the Monte Carlo channel realizations,

$$SE_k^i = \tau_i (1 - \tau_{\text{sig},i}) \bar{x}_i \mathbb{E} \left(\frac{1}{Q} \sum_{q=1}^Q \log_2 (1 + \text{SINR}_k^i[q]) \right) \quad (18)$$

with $i \in \{\text{DL}, \text{UL}\}$, the prelog is the data fraction (3) of the frame, since reference signalling and the deactivated symbols carry no payload. The delivered sum rates can then be written as $R_i = \tilde{B} \sum_k SE_k^i$, with the effective bandwidth $\tilde{B} = 0.9 B$.

III. POWER MODEL DESCRIPTION

We first describe the consumption of a distributed network and outline the frame averaging, the functional split and the placement of the MIMO processing (Section III-A). Next, the two components that are specific to D-MIMO, i.e., the fronthaul and the synchronization of physically separate nodes (Section III-B) are described. Finally we describe the parametric model of the AP consumption, which is based on [1] (Section III-C).

TODO if no space can cut this cause a bit redundant. Also check if we keep load=1 or if we vary it in some sims The topology, the indices l, m, k, q , the per-AP budget $\rho_{\max}$ and radiated power $P_{T,l}$, the frame structure and the delivered rates $R_{\text{DL}}, R_{\text{UL}}$ are those of Section II, and $i \in \{\text{DL}, \text{UL}\}$ indexes denote the transmission directions. The APs are fully digital, so each carries M radio frequency (RF) chains. Every AP is assumed active and serves every user, and both directions are evaluated at full physical resource load, $\bar{x}_{\text{DL}} = \bar{x}_{\text{UL}} = 1$.

A. Network Power Consumption

The network consumption contains the L APs, the L fronthaul links that connect them to the CPU, and the CPU itself,

$$P_{\text{net}} = \sum_{l=1}^L \left(P_{\text{AP},l} + \frac{\overline{P_{\text{FH},l}}}{\eta_{\text{FH},\text{sc}}} \right) + \frac{\overline{P_{\text{CPU}}}}{\eta_{\text{CPU},\text{sc}}}, \quad (19)$$

where an overbar denotes a quantity averaged over one frame, as outlined in Section III-A1, and $\eta_{\text{FH},\text{sc}}, \eta_{\text{CPU},\text{sc}} \in (0, 1]$ are the supply-and-cooling efficiencies of the fronthaul transceivers and of the central unit. Each active AP retains the three-block structure of the co-located model from [1] and adds a constant synchronization consumption P_{sync} , detailed in Section III-B2,

$$P_{\text{AP},l} = \frac{\overline{P_{\text{dig},l}}}{\eta_{\text{dig},\text{sc}}} + \frac{\overline{P_{\text{ana},l}}}{\eta_{\text{ana},\text{sc}}} + \frac{\overline{P_{\text{PA},l}}}{\eta_{\text{PA},\text{sc}}} + \frac{P_{\text{sync}}}{\eta_{\text{sync},\text{sc}}}, \quad (20)$$

with the digital, analog and amplifier blocks detailed in Section III-C. The co-located baseline is recovered by setting $L = 1$, $M = M_{\text{ant}}$ and dropping the synchronization, fronthaul and CPU terms.

TABLE I: Node at which each digital operation runs. The uplink combiner follows the downlink precoder. Channel estimation is listed for completeness but is not priced, since perfect CSI is assumed.

Operation	S1	S2	S3
Encoding / decoding	CPU	CPU	CPU
Channel estimation	CPU	CPU	AP
Precoder computation	CPU	CPU	AP
Precoder application	CPU	AP	AP
OFDM, predistortion, BB filter	AP	AP	AP

1) *Frame structure and operating-mode averaging*: The overbars in (19) and (20) are averages over one frame, and every block of the power model is averaged over a frame. The frame structure is defined in Section II. The frame timing is shared by the whole network: under TDD the APs must switch direction together. In micro-sleep and idle a component drops to a fraction $\delta_c^\mu, \delta_c^0 \in [0, 1]$ of a base level P^0 . For a component c with working power P^a during data and P^s during signalling, the frame-averaged consumption of direction i is

$$\begin{aligned} \mathcal{A}_i^c(P^a, P^s, P^0) &= \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) P^a + \tau_i \tau_{i,\text{sig}} P^s \\ &\quad + (1 - \bar{x}_i) \tau_i (1 - \tau_{i,\text{sig}}) \delta_c^\mu P^0 \\ &\quad + (1 - \tau_i) \delta_c^0 P^0. \end{aligned} \quad (21)$$

For the digital and analog blocks the three levels coincide, $P^a = P^s = P^0 = P$, and we write $\mathcal{A}_i^c(P) \triangleq \mathcal{A}_i^c(P, P, P)$; only the amplifier distinguishes them. Collecting the terms proportional to $\bar{x}_i$ separates each averaged consumption into a load-dependent and a load-independent part, which is what allows the consumption to be attributed to the delivered traffic.

2) *Functional split*: A single co-located site runs every digital operation in one place. A distributed network does not, and where each operation runs determines both the digital consumption and the fronthaul load. We consider the three splits of Table I. Split S1 forwards precoded samples: the CPU encodes, computes the centralized precoder and applies it, so AP l receives the M streams of frequency-domain samples. Split S2 forwards precoder coefficients and data: the CPU encodes and computes the precoder, but AP l applies it locally. Split S3 is fully local operation: the CPU only encodes, and each AP computes and applies its own precoder from local CSI. Splits S1 and S2 realize the same centralized precoder and therefore deliver the same rate; they differ only in where the work runs and in what crosses the fronthaul. S3 delivers a lower rate due to local processing.

3) *MIMO processing and its placement*: Following [1], a digital block consumes a static part plus its complex operations per sample Ξ evaluated at the constellation sampling rate $f_{s,\text{I}}$ and divided by the computational efficiency η of the implementation,

$$P_{\text{MIMO}} = \begin{cases} \bar{M} P_s + \frac{f_{s,\text{I}}}{\eta} \Xi, & \Xi > 0, \\ 0, & \Xi = 0, \end{cases} \quad (22)$$

with η the computational efficiency of the precoder in the downlink and of the combiner in the uplink, P_s the static

TABLE II: Operations needed to *compute* the precoders and combiners of Section II-C, per subcarrier and per coherence block, derived in ???. Dividing by v_{coh} gives the per-sample count Ξ_{comp} of (23). Applying the result costs a further $2K$ operations per sample and per processed antenna, for every scheme. **TODO maximum ratio (MR) is in the rate model but is not yet described in Section II-C; drop that row if it stays out of the paper.**

Scheme	Node	$v_{\text{coh}} \Xi_{\text{comp}}$
MR	any	0
ZF, RZF, MMSE	CPU	$\frac{K^3}{3} + 3K^2LM + KLM$
L-RZF, L-MMSE	AP l	$\frac{M^3}{3} + 3M^2K + MK$

power of one field-programmable gate array (FPGA) and $\bar{M}$ the number of FPGAs. Following [1], one FPGA per group of 32 RF chains: $\bar{M} = \lceil M/32 \rceil$ is used at an AP and $\bar{M} = \lceil LM/32 \rceil$ at the CPU, which processes the collective array. When considering S1, the AP has no precoder block at all, so $\Xi = 0$ and the AP has no idle FPGA.

Computing the precoder or combiner is considered separately from applying it to the signal, because the split moves these two operations between the AP and the CPU. Applying an already computed precoder or combiner to K UEs costs $2K$ operations per sample and per processed antenna whatever the scheme, hence $2KLM$ at the CPU and $2KM$ at one AP. Computing the precoder/combiner requires the scheme-dependent Ξ_{comp} of Table II, once per coherence block, and this cost is therefore spread over the v_{coh} samples of that block. We take the frame of Section II to span one coherence block, so the precoder is recomputed once per frame and the reference signalling fraction $\tau_{i,\text{sig}}$ is the share of that same block spent on pilots: one fading timescale governs both the precoder amortisation here and the signalling overhead of (18). The splits distribute the two counts over the two nodes as

$$\Xi_i^{\text{CPU}} = \iota_{S1} 2KLM + \iota_{S1,S2} \iota_{\text{DL}} \Xi_{\text{comp}}^{\text{cen}}, \quad (23)$$

$$\Xi_{i,l}^{\text{AP}} = \iota_{S2,S3} 2KM + \iota_{S3} \Xi_{\text{comp}}^{\text{loc}}, \quad (24)$$

where $\iota_{\bullet} \in \{0,1\}$ equals one for the split or direction named in its subscript. The centralized factorization is charged to the downlink only, since the ZF combiner of Section II-C is the same matrix as the ZF precoder and TDD reciprocity lets one factorization serve both directions. The local pair shares no factorization, because the uplink weights the local Gram matrix by the transmit powers $\mathbf{P}$ in (16) while the downlink does not, so under S3 the cost is charged in each direction.

4) *Central unit*: Channel encoding and decoding stay at the CPU under every split, since that is where the payload enters and leaves the network, and are driven by the delivered network sum rates R_{DL} and R_{UL} through the encoder and decoder models $P_{\text{enc}}, P_{\text{dec}}$ of [1]. With the frame-averaging operator $\mathcal{A}_i$ of (21), the central unit consumes

$$\begin{aligned} \overline{P_{\text{CPU}}} &= P_{\text{CPU},0} + \mathcal{A}_{\text{DL}}(P_{\text{enc}} + P_{\text{MIMO,DL}}^{\text{CPU}}) \\ &+ \mathcal{A}_{\text{UL}}(P_{\text{dec}} + P_{\text{MIMO,UL}}^{\text{CPU}}), \end{aligned} \quad (25)$$

with $P_{\text{CPU},0}$ the always-on consumption of the central unit. It does not cover the baseband hardware: the FPGAs on which the collective array is processed are already charged through

the $\bar{M}P_s$ term of (22), with $\bar{M} = \lceil LM/32 \rceil$, and the encoder and decoder carry their own static power inside P_{enc} and P_{dec} . $P_{\text{CPU},0}$ is therefore only what remains, the chassis and the infrastructure of the central unit, together with the shared reference clock of Section III-B2.

B. Fronthaul and Synchronization

1) *Fronthaul*: Following the cell-free power model of [5], the link serving AP l consumes a traffic-independent part covering the optical transceivers at both ends and a part proportional to the rate it carries,

$$P_{\text{FH},l} = P_{\text{FH},0} + \Pi_{\text{FH}} R_{\text{FH},l}, \quad (26)$$

with Π_{FH} in W per bit/s. What $R_{\text{FH},l}$ contains is set by the split of Section III-A2. Let b_{FH} be the number of bits per real sample, so a complex sample costs $2b_{\text{FH}}$ bits. Under S1 the link carries M streams of frequency-domain samples at the constellation rate in both directions and in both the data and the signalling mode,

$$R_{\text{FH},l}^{\text{S1}} = 2b_{\text{FH}} M f_{s,\text{I}}. \quad (27)$$

Under S2 the downlink carries the delivered payload plus the precoding coefficients, an $M \times K$ complex matrix per subcarrier refreshed once per coherence block,

$$R_{\text{FH},l}^{\text{S2,DL}} = R_{\text{DL}} + \frac{f_{s,\text{I}}}{v_{\text{coh}}} 2b_{\text{FH}} M K, \quad (28)$$

where R_{DL} appears in full because every AP serves every user. Under S3 only the payload is sent over the fronthaul links, $R_{\text{FH},l}^{\text{S3,DL}} = R_{\text{DL}}$. In the uplink, linear combining is distributive over the APs, $\hat{s}_k = \sum_l \mathbf{v}_{kl}^H \mathbf{y}_l$, so an AP holding its combining coefficients forwards K scalar partial sums instead of M sample streams under both S2 and S3,

$$R_{\text{FH},l}^{\text{S2,UL}} = R_{\text{FH},l}^{\text{S3,UL}} = 2b_{\text{FH}} K f_{s,\text{I}}. \quad (29)$$

The signalling mode separates the centralized splits from the local one. Centralized operation needs the CPU to see the raw received pilots in order to estimate the collective channel, so during the uplink signalling phase the link carries $2b_{\text{FH}} M f_{s,\text{I}}$ under S1 and S2 alike, whereas under S3 the pilots are consumed locally and the signalling load vanishes. Centralization therefore imposes a sample-level fronthaul during pilot transmission independent of the data-phase split.

The link is bidirectional and stays up across the whole frame, so its static term is charged once per link with a single duty cycle rather than once per direction. With

$$\theta_l = \sum_{i \in \{\text{DL}, \text{UL}\}} \tau_i [\tau_{i,\text{sig}} + (1 - \tau_{i,\text{sig}}) \bar{x}_i] \quad (30)$$

the fraction of the frame in which the link is in a working mode, and δ_{FH}^μ the reduction factor of an idle transceiver,

$$\overline{P_{\text{FH},l}} = [\theta_l + (1 - \theta_l) \delta_{\text{FH}}^\mu] P_{\text{FH},0} + \Pi_{\text{FH}} \overline{R_{\text{FH},l}}. \quad (31)$$

The averaged fronthaul rate reuses (21) with a zero base level, since an idle link carries no traffic,

$$\overline{R_{\text{FH},l}} = R_{\text{FH},l}^{\text{payload}} + \sum_{i \in \{\text{DL}, \text{UL}\}} \mathcal{A}_i(R_{\text{FH},l}^i, R_{\text{FH},l}^{i,\text{sig}}, 0), \quad (32)$$

where $R_{\text{FH},l}^{\text{payload}}$ is the delivered downlink payload of the data-sharing splits S2 and S3 and is excluded from the averaging, the remaining arguments then being peak rates. The reason is that the map from rate to power in (26) is linear and $\mathcal{A}_i$ with a zero base is linear in its arguments, while the rate model already delivers R_{DL} with its prelog applied. Charging the delivered rate directly is thus equivalent to frame-averaging the corresponding peak rate, and doing both would count the frame structure twice. For the sample-forwarding split S1 the situation is the opposite: (27) is a hardware constant with no dependence on the delivered rate, and the frame structure is then the only mechanism that reduces the fronthaul load.

Remark 1 (Fronthaul ceiling on energy efficiency). *Under a data-sharing split (S2,S3) the same payload is duplicated to every AP, so the network fronthaul load is $L R_{\text{DL}}$. Keeping only this term in (19) bounds the downlink energy efficiency by $1/(\Pi_{\text{FH}} L)$, independently of the bandwidth, the transmit power and the precoder. With $L = 32$ and $\Pi_{\text{FH}} = 0.25 \text{ W per Gbit/s}$ the network cannot exceed 125 Mbit J^{-1} however well everything else is designed, which is why the serving cluster size is the dominant lever on cell-free energy efficiency.*

2) *Synchronization*: Coherent joint transmission across physically separate nodes requires the APs to share a frequency and phase reference and requires the TDD chains to be reciprocity-calibrated. Whether the reference is distributed over the fronthaul, over a dedicated timing network, or over the air, the AP must hold it whether or not the frame is carrying data. It is therefore considered as the constant P_{sync} of (20), entirely load-independent and outside the frame averaging, with its own supply-and-cooling efficiency $\eta_{\text{sync,sc}}$.

What P_{sync} covers is the AP-side reference hardware only, that is the disciplined oscillator and the loop that slaves it to a reference. A shared master clock is not a per-AP cost and is counted in $P_{\text{CPU},0}$ instead. The over-the-air part of reciprocity calibration is frame time and CPU computation rather than a continuous per-AP power, and is not modelled.

C. Access Point and Co-located Power Consumption

The three frame-averaged blocks of (20) come from the FR3 BS model of [1], evaluated once per AP. A distributed AP carries few antennas, so hybrid beamforming has nothing to amortise and the array is fully digital, $M_{\text{RF}} = M_{\text{ant}} = M$ with no phase shifters. Substituting this into the analog and amplifier assemblies of [1] reduces them term for term to the per-AP expressions below. The digital block cannot be reused directly from [1] because the encoder and decoder move to the CPU and the MIMO processing terms depend on the functional split.

1) *Power amplifiers*: The amplifiers transmit only in the downlink. With output power p , the instantaneous consumption of one amplifier is

$$P_{\text{PA}}(p) = \frac{1}{\eta_{\text{PA,max}}} [\xi P_{\text{max}} + (1 - \xi) P_{\text{max}}^{1-\alpha} p^\alpha], \quad (33)$$

with P_{max} the maximum output power at a fixed back-off, $\eta_{\text{PA,max}}$ the maximum efficiency, $\xi \in [0, 1]$ the static-to-

dynamic ratio and $\alpha \in [0, 1]$ the efficiency exponent. Each AP is an independent transmitter with its own budget ρ_{max} , so the single site power of [1] becomes the vector of radiated powers $P_{T,l}$ of (11), supplied by the rate model, which the two operation modes fill in differently: distributed operation meets the constraint with equality at every AP, $P_{T,l} = \rho_{\text{max}}$, whereas centralized operation rescales all APs by one common factor and brings only the busiest of them to its budget. Assuming the AP power is shared equally across its antennas, $P_{a,l} = P_{T,l}/M$, and

$$\overline{P_{\text{PA},l}} = M \mathcal{A}_{\text{DL}}^{\text{PA}}(P_{\text{PA}}(P_{a,l}), P_{\text{PA}}(\zeta_{\text{sig}} P_{\text{max}}), P_{\text{PA}}(0)), \quad (34)$$

with ζ_{sig} the fraction of P_{max} used during signalling. The equal split is not exact, since the per-antenna powers differ for any non-isotropic precoder, but because $\alpha \in [0, 1]$ makes $P_{\text{PA}}(\cdot)$ concave, Jensen's inequality gives $\frac{1}{M} \sum_m P_{\text{PA}}(P_{a,lm}) \leq P_{\text{PA}}(P_{a,l})$: it is a conservative upper bound, tight when the precoder loads an AP uniformly.

Remark 2 (Amplifier sizing). *The relation between P_{max} and the budget decides how the static term scales with the deployment and must be stated. We size each amplifier for its share of the budget it is actually given, $P_{\text{max}} = \rho_{\text{max}}/M$ at an AP and $P_{\text{max}} = P_{\text{TX}}/(LM)$ at the co-located site. Since the comparison holds the total transmit power fixed, $\rho_{\text{max}} = P_{\text{TX}}/L$, the two deployments then carry amplifiers of identical rating and differ only in how many sites they sit on. Substituting (33) into (34) makes the data-mode consumption of an AP independent of M ,*

$$M P_{\text{PA}}(P_{a,l}) = \frac{\rho_{\text{max}}}{\eta_{\text{PA,max}}} \left[\xi + (1 - \xi) \left(\frac{P_{T,l}}{\rho_{\text{max}}} \right)^\alpha \right], \quad (35)$$

so the amplifiers see the precoders of Section II-C only through the fraction $P_{T,l}/\rho_{\text{max}} \in [0, 1]$ of the budget that AP l radiates, and the network static floor is $L\xi\rho_{\text{max}}/\eta_{\text{PA,max}} = \xi P_{\text{TX}}/\eta_{\text{PA,max}}$, the same for both deployments at equal total transmit power. The per-AP constraint then costs the network twice under centralized operation: only part of the budget is radiated, while the static floor is paid in full. The convention also fixes the operating budget, since P_{max} is a device rating rather than a free parameter: at the FR3 amplifier of [1] the network budget is $P_{\text{TX}} = LM P_{\text{max}}$, which is how Table III is set.

2) *Analog front end*: The analog block comprises, per RF chain, the digital-to-analog converters (DACs) in the downlink or analog-to-digital converters (ADCs) in the uplink, RF filters, mixers and the low-noise amplifiers, with the local oscillator shared across the chains of one node. Specialising [1] to a fully digital array without phase shifters,

$$P_{\text{ana,DL},l} = P_{\text{LO}} + M(2P_{\text{DAC}} + 2P_{\text{filt,RF}} + 2P_{\text{mix}}), \quad (36)$$

$$P_{\text{ana,UL},l} = P_{\text{LO}} + M(2P_{\text{ADC}} + 2P_{\text{filt,RF}} + 2P_{\text{mix}} + P_{\text{LNA}}), \quad (37)$$

the factor two accounting for the in-phase and quadrature branches, and

$$\overline{P_{\text{ana},l}} = \mathcal{A}_{\text{DL}}^{\text{ana}}(P_{\text{ana},\text{DL},l}) + \mathcal{A}_{\text{UL}}^{\text{ana}}(P_{\text{ana},\text{UL},l}) + P_{\text{ana},\text{misc}}. \quad (38)$$

This is the co-located analog block term for term, with no distributed addition: the synchronization of Section III-B2 sits outside it, in (20). The local oscillator is the first term to change character: a co-located array shares one across all M_{ant} chains, whereas the distributed network pays $L P_{\text{LO}}$.

3) *Digital signal processing*: Every AP performs the OFDM modulation and demodulation, the predistortion of its own amplifiers and the baseband filtering whatever the split, so with the component models P_{IFFT} , P_{DPD} and $P_{\text{filt,BB}}$ of [1] evaluated at M chains,

$$P_{\text{dig},\text{DL},l} = P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filt,BB}} + P_{\text{MIMO},\text{DL},l}, \quad (39)$$

$$P_{\text{dig},\text{UL},l} = P_{\text{IFFT}} + P_{\text{filt,BB}} + P_{\text{MIMO},\text{UL},l}, \quad (40)$$

with $\overline{P_{\text{dig},l}} = \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{dig},\text{DL},l}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dig},\text{UL},l})$. The encoder and decoder are absent by construction, which is the structural difference from the single-site model, and only the MIMO term of (22) depends on the split.

Remark 3 (Loss of static-power amortisation). *The precoder, combiner and baseband filter share a static power P_s per FPGA, one per group of 32 chains. A co-located array of M_{ant} antennas amortises that power over 32 chains at a time, whereas a network of L APs with $M < 32$ antennas each pays it L times. At $L = 32$ and $M = 4$ the network runs 32 FPGAs where a co-located array of the same 128 antennas runs 4. Chopping a large array into many small ones leaves the per-chain digital costs untouched but destroys the sharing of the fixed ones, and this is a general feature rather than an artefact of the value 32.*

4) *Co-located baseline*: The reference deployment is the unmodified model of [1] evaluated at a single site: one fully digital array of $M_{\text{ant}} = LM$ antennas radiating the full budget P_{TX} , one shared local oscillator, no fronthaul, no central unit and no synchronization power, with the complete digital chain (encoder, precoder, OFDM, predistortion and baseband filter in the downlink; decoder, combiner, OFDM and baseband filter in the uplink) at that one site.

D. Delivered Rate and Energy Efficiency

The delivered sum rates R_{DL} and R_{UL} come from the rate model of Section II with the data-fraction prelogs of (18) already applied, and the network energy efficiency is the delivered rate per consumed watt,

$$\text{EE}_{\text{net}} = \frac{R_{\text{DL}} + R_{\text{UL}}}{P_{\text{net}}} \quad [\text{bit/J}], \quad (41)$$

which reduces to the single-site energy efficiency of [1] for the co-located baseline. Because a distributed and a co-located deployment cover the same area with the same hardware count, the comparison is made at equal LM and equal total transmit power.

TABLE III: Deployment, channel and frame parameters of Section II.

Symbol	Description	Value
<i>Topology</i>		
L	Number of APs, swept	$1, 2, \dots, 128$
M	Antennas per AP, LM/L	$128, \dots, 1$
LM	Total antennas, both deployments	128
K	Single-antenna UEs	20
Q	Data subcarriers, $\tilde{B}/\Delta_f$	3000
D	Coverage-area side	200 m
h_{AP}	AP height [3]	10 m
h_{UE}	UE height [3]	1.5 m
<i>Radio</i>		
f_c	Carrier frequency	10 GHz
B	Bandwidth [1]	400 MHz
Δ_f	Subcarrier spacing [1]	120 kHz
$\tilde{B}$	Effective bandwidth, $0.9B$	360 MHz
F	Receiver noise figure	8 dB
σ^2	Noise power, from Section II-B	-80.0 dBm
<i>Transmit power and power control</i>		
P_{TX}	Total budget, $LM P_{\text{max}}$	12.8 W
ρ_{max}	Per-AP budget, P_{TX}/L	0.4 W at $L = 32$
p_{max}	Per-UE uplink budget	0.1 W
v	Downlink fractional exponent	0.5
v_{ul}	Uplink fractional exponent	-0.5
λ	Local RZF loading	σ^2
<i>Frame</i>		
τ_{DL}	Downlink share of the frame	0.75
τ_{UL}	Uplink share, $1 - \tau_{\text{DL}}$	0.25
$\tau_{\text{DL},\text{sig}}$	Downlink signalling fraction	1/14
$\tau_{\text{UL},\text{sig}}$	Uplink signalling fraction	1/14
$\bar{x}_{\text{DL}}, \bar{x}_{\text{UL}}$	Resource load	1
v_{coh}	Samples per coherence block	200
<i>Monte Carlo</i>		
	Channel realizations per point	20
Q_{MC}	Subcarriers evaluated per realization	64
	Channel model [3], [4]	38.901 UMI

IV. NUMERICAL EVALUATION

A. Parameters

The evaluation parameters are split up into three groups. The deployment, channel and frame of Section II are collected in Table III. The hardware constants of the AP and of the co-located baseline are inherited from the FR3 BS model of [1] and are listed in Table IV. The quantities that exist only in a distributed network, that is the fronthaul, the central unit and the per-AP synchronization, are collected in Table V.

Note that the number of data subcarriers is set to $Q = 3000$ for the power model. However, the rate model instead evaluates the subcarrier average of (18) on 64 subcarriers and scales the resulting spectral efficiency by $\tilde{B}$, as evaluating all 3000 subcarriers would consume unnecessary compute time.

B. Evaluation Scenarios

We consider four evaluations that use the parameters of Tables III to V. All of them hold the total antenna count at $LM = 128$ and the total transmit budget at P_{TX} , the two deployments cover the same area with the same radiating

TABLE IV: Hardware constants of the AP and of the co-located baseline, inherited from the FR3 BS model of [1] and applied to both deployments.

Symbol	Description	Value	Symbol	Description	Value
<i>Supply and cooling</i>			<i>Analog front end</i>		
$\eta_{\text{dig,sc}}$	Digital block	0.8	b_{DAC}	DAC effective bits	8
$\eta_{\text{ana,sc}}$	Analog block	0.8	f_{DAC}	DAC sampling frequency	5 GHz
$\eta_{\text{PA,sc}}$	Amplifier block	0.8	$\Xi_{\text{DAC},1}$	DAC static coefficient	$1.5 \times 10^{-5} \text{ W}$
<i>Power amplifier</i>			$\Xi_{\text{DAC},2}$	DAC dynamic coefficient	$1.5 \times 10^{-12} \text{ J bit}^{-1}$
ξ	Static-to-dynamic ratio	0.1	b_{ADC}	ADC effective bits	8
α	Efficiency exponent	0.75	f_{ADC}	ADC sampling frequency	5 GHz
$\eta_{\text{PA,max}}$	Maximum efficiency	0.15	W_{ADC}	ADC Walden figure of merit	70 fJ
P_{max}	Rating, ρ_{max}/M	0.1 W	Ξ_{LNA}	low-noise amplifier (LNA), $P_{\text{LNA}} = \Xi_{\text{LNA}} B$	$2.7 \times 10^{-11} \text{ W Hz}^{-1}$
ζ_{sig}	Signalling power fraction	1/12	Ξ_{mix}	Mixer, $P_{\text{mix}} = \Xi_{\text{mix}} f_c$	$2.5 \times 10^{-13} \text{ W Hz}^{-1}$
<i>Digital processing</i>			P_{LO}	Local oscillator	40 mW
Q_{IFFT}	inverse fast Fourier transform (IFFT) size	4096	$P_{\text{filt,RF}}$	RF filter	5 mW
μ	Sampling-frequency factor	0.9	$P_{\text{ana,misc}}$	Miscellaneous analog	0 W
$f_{s,I}$	Constellation rate, μB	360 MHz	<i>Computational efficiencies [complex GOPS/W]</i>		
$f_{s,II}$	Baseband rate, $Q_{\text{IFFT}} \Delta_f$	491.52 MHz	$\eta_{\text{enc}}, \eta_{\text{dec}}$	Encoder, decoder (ASIC)	2000
Ξ_{DPD}	Predistortion, ops per sample	50	$\eta_{\text{IFFT}}, \eta_{\text{DPD}}$	IFFT, predistortion (ASIC)	2000
$n_{\text{filt,BB}}$	Baseband-filter taps	20	η	Precoder, combiner (FPGA)	200
$o_{\text{filt,BB}}$	Baseband oversampling	4	$\eta_{\text{filt,BB}}$	Baseband filter (FPGA)	200
P_s	Static power per FPGA	1 W			
	RF chains per FPGA	32			
<i>Reduction factors (micro-sleep / idle)</i>					
$\delta_{\text{dig}}^\mu, \delta_{\text{dig}}^0$	Digital	0.5, 0.25			
$\delta_{\text{ana}}^\mu, \delta_{\text{ana}}^0$	Analog	0.75, 0.5			
$\delta_{\text{PA}}^\mu, \delta_{\text{PA}}^0$	Amplifier	0.5, 0.25			

TABLE V: Parameters specific to the distributed network.

Symbol	Description	Value	Source
<i>Fronthaul</i>			
$P_{\text{FH},0}$	Traffic-independent power	0.825 W	[5]
Π_{FH}	Traffic-dependent coefficient	0.25 W/Gbit/s	[5]
b_{FH}	Bits per real sample	8	
δ_{FH}^μ	Idle transceiver reduction	0.25	assumed
$\eta_{\text{FH,sc}}$	Supply and cooling	0.8	assumed
<i>Central unit</i>			
$P_{\text{CPU},0}$	Always-on consumption	0 W	assumed
$\eta_{\text{CPU,sc}}$	Supply and cooling	0.8	assumed
	RF chains per CPU FPGA	32	[1]
<i>Synchronization</i>			
P_{sync}	Per-AP reference hardware	0.1 W	
$\eta_{\text{sync,sc}}$	Supply and cooling	0.8	assumed

hardware and the same radiated power, and differ only in how finely that hardware is spread. Distributing therefore never adds antennas, it only redistributes them. The amplifier rating $P_{\text{max}} = \rho_{\text{max}}/M = P_{\text{TX}}/(LM)$ is the same at every point of every evaluation, so the amplifiers are physically identical hardware throughout and any change in the amplifier consumption comes from utilisation alone. **TODO this probably captures enough so that remark 2 could be shortened or removed** The first evaluation considers the delivered rate against the consumed power, by sweeping P_{TX} , with the amplifiers re-sized at each point so that every point on the curve is a deployment dimensioned for the budget it is given. It

is reported for the co-located baseline and for the three splits of Table I, at $L \in \{32, 64, 128\}$, that is at $M \in \{4, 2, 1\}$ antennas per AP. The second evaluation varies L over $2, 4, \dots, 128$ at fixed P_{TX} and reports P_{net} , again for the three splits. The co-located baseline of Section III-C is drawn alongside it at $L = 1$. The third evaluation considers the same sweep as evaluation two but looks at the contributions of (19) and (20), namely the amplifier, digital, analog, fronthaul, central-unit and synchronization blocks. The fourth evaluation reports the energy efficiency (41) over the same sweep. Because the delivered rate is not constant along it, P_{net} alone cannot say whether distributing is beneficial. The fronthaul ceiling $1/(\Pi_{\text{FH}}L)$ of Remark 1 is drawn alongside the data-sharing splits.

C. Evaluation 1: Delivered Rate against Consumed Power

Figure 1 illustrates the delivered rate against the consumed power, when sweeping the transmit power. Distributing the array enables a higher rate but results in a higher power consumption. At the nominal budget $P_{\text{TX}} = 12.8 \text{ W}$ and $L = 32$, centralized D-MIMO delivers 69.7 Gbit s^{-1} against 44.4 Gbit s^{-1} for the co-located site, a factor 1.57, but draws 470 W against 188 W . The rate gain is overwhelmingly in the uplink, 14.9 Gbit s^{-1} against 4.9 Gbit s^{-1} , which is macro diversity working as expected: every UE is close to some AP, whereas a single central array serves the edge of the coverage area over the full path loss.

The shape difference in the curves of the different deployments is significant. The co-located curve is the only one

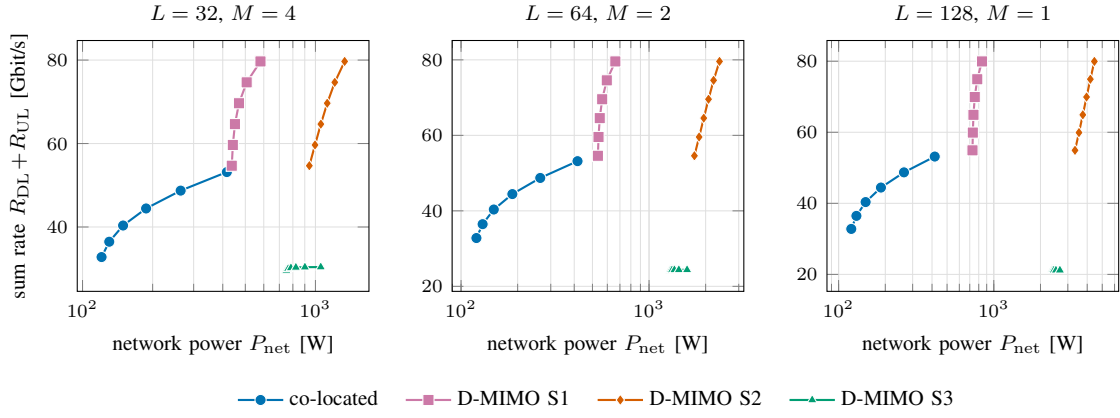


Fig. 1: Delivered sum rate against network power consumption, at $LM = 128$ antennas and $K = 20$ UEs. Each curve is obtained by sweeping the total transmit budget over $P_{TX} \in \{1.6, 3.2, 6.4, 12.8, 25.6, 51.2\}$ W, increasing upward, with the amplifiers re-sized at every point.

that moves substantially along the power axis, from 121 W to 417 W across a 32-fold increase in the power budget. The distributed curves S1 and S2 are close to vertical: the rate climbs while P_{net} marginally increases. This highlights the fact that the distributed network carries a large consumed power overhead due to the fronthaul and distributed processing. Additionally, split S3 has a relatively flat rate when the transmit budget increases. Since local processing is interference-limited rather than noise-limited, raising every AP's power raises the interference with the signal and thus results in a stable SINR.

The three distributed curves have a different power consumption, mainly due to the fronthaul requirements. A link operated under S1 carries M quantized sample streams over the fronthaul in either direction by (27), so the network transports LM streams. A link operated under S2 or S3 carries the traffic expressed in the user domain instead: K uplink partial sums per AP according to (29), together with the downlink payload duplicated to every AP. Moving the precoder and the combiner out to the APs therefore converts antenna-domain streams into user-domain streams, and that conversion compresses the link only when an AP has more antennas than served UEs. Considering the scenario of $K = 20$ UEs and $M = \{4, 2, 1\}$ the fronthaul consumption of S2 and S3 are much higher than the one of S1. Split S2 is accordingly the most expensive of the three and worsens fastest with L , from 1124 W to 3962 W between $L = 32$ and $L = 128$ at the nominal budget, while delivering exactly the rate of S1. S3 has a slightly lower consumption as compared to S2, since local processing delivers a lower rate, and as such requires a lower fronthaul rate.

D. Evaluation 2: Network Power against the Number of APs

Figure 2 reports network power when L is varied. No distributed deployment is cheaper than the co-located baseline anywhere on the sweep. The cheapest distributed point of the whole figure is S3 at $L = 2$, at 238 W against 187 W for the single site, and the gap widens monotonically from there. Since every point radiates the same budget from the same 128

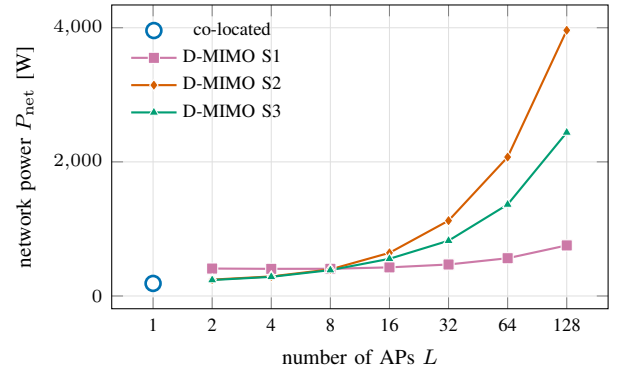


Fig. 2: Network power consumption against the number of APs, at the fixed total budget $P_{TX} = 12.8$ W, the fixed antenna total $LM = 128$ and $K = 20$ UEs.

antennas, the entire difference is the cost of separating them, i.e., fronthaul and processing cost.

E. Evaluation 3: Where the Power Goes

Figure 3 breaks down the total network power of Figure 2 into the respective blocks.

The power amplifier (PA) consumption under local processing (S3) is 76.3 W at every L , exactly the co-located value: each AP transmits $P_{T,l}/\rho_{\text{max}} = 1$ and the sizing convention of Remark 2 makes the network amplifier consumption independent of how the array is cut. Under centralized precoding (S1, S2) it instead falls monotonically, from 64.9 W at $L = 2$ to 28.5 W at $L = 128$. The measured mean utilisation $P_{T,l}/\rho_{\text{max}}$ falls from 0.78 to 0.19 over the same range, which accounts for it through (35): the per-AP constraint of (12) binds at the most loaded AP and scales the joint precoder for all of them, so the rest under-radiate, and the finer the array is cut the wider that spread becomes. Notably, under centralized processing, distributing the array reduces the amplifier consumption, however, the global consumption P_{net} still rises.

The digital block confirms Remark 3, and it separates relocating the processing from duplicating it. At $L = 2$ the

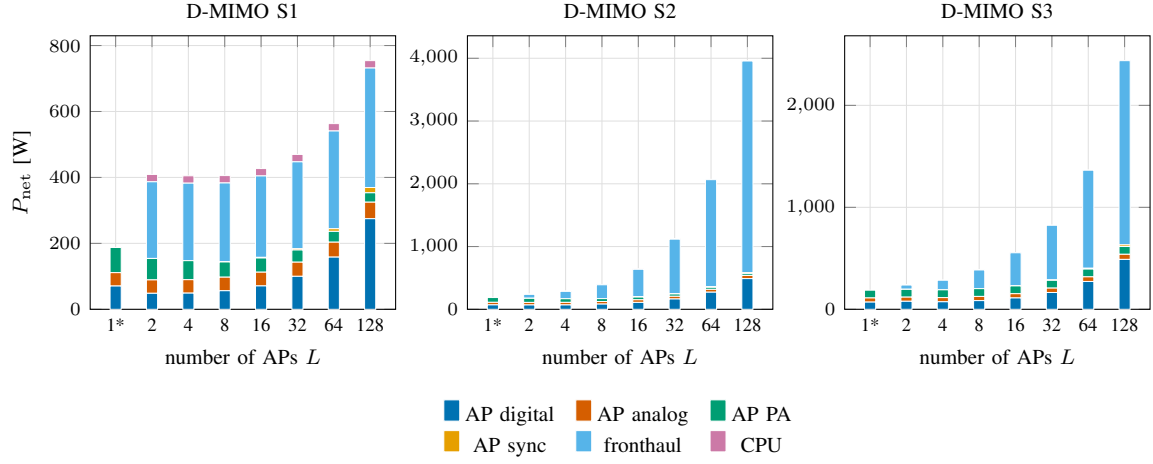


Fig. 3: The same sweep as Figure 2, resolved into the blocks of (19) and (20). The three panels do not share a vertical scale.

S1 network spends 48.6 W at the APs and 22.3 W at the central unit, so 70.9 W in total against 70.6 W for the co-located site: moving the precoder and combiner off the APs costs essentially nothing. The growth that follows is the loss of sharing. By $L = 128$ the same two terms are 274.8 and 22.4 W, a factor 4.2 over the baseline, while the central unit itself has not moved at all, since it runs $\lceil LM/32 \rceil = 4$ FPGAs whatever L is. S2 and S3 make every AP run its own precoder or combiner rather than only its baseband filter, this results in an increased consumption: 489 W of AP digital at $L = 128$, a factor 7.0 including the central unit.

The fronthaul has the biggest impact. It grows for two different reasons. Under S1 it is nearly stable when L varies, i.e., 232 W at $L = 2$ against 362 W at $L = 128$, and the traffic dependent term does not increase at all: it is 230 W for both values of L . The 130 W of growth is the traffic-independent $P_{FH,0}/\eta_{FH,sc}$ paid once per link, which the 126 additional links account for. Under S2 and S3 that same static term is small compared to the traffic dependent consumption, 132 W out of the 3373 and 1807 W they reach at $L = 128$, where the fronthaul is 85 % and 74 % of everything the network consumes.

The remaining two blocks are small enough to be dismissed. Synchronization is exactly linear in L and reaches 16 W at $L = 128$. The analog block grows by 75 mW per additional AP, its own local oscillator, so it moves only from 40.6 to 50.1 W. Neither the reference distribution nor the loss of the shared oscillator is a reason not to distribute.

In conclusion, the cost of distributing an array is transport and per-node fixed overhead, not radiated power: at $L = 128$ under S1 the fronthaul and the digital blocks add 570 W to a 188 W baseline.

F. Energy Efficiency

Figure 4 reports the energy efficiency against the number of APs. For the considered network distributing is not beneficial: the co-located baseline reaches 237.0 Mbit J⁻¹ and no distributed configuration of any split at any L comes within

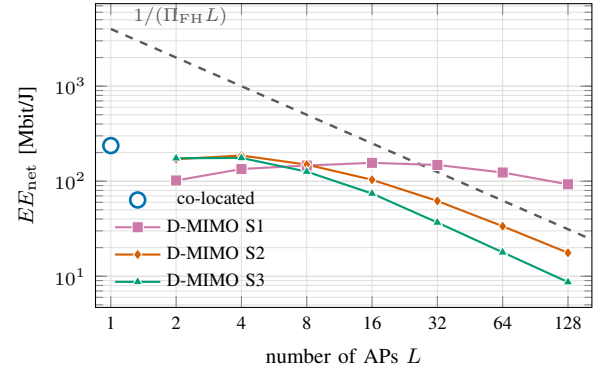


Fig. 4: Network energy efficiency (41) over the sweep of Figure 2, at $P_{TX} = 12.8$ W, $LM = 128$ antennas and $K = 20$ UEs. Both axes are logarithmic, which makes the ceiling $1/(\Pi_{FH}L)$ of Remark 1 the straight dashed line. That ceiling bounds the data-sharing splits S2 and S3, whose fronthaul carries the payload duplicated to every AP. It does not bound S1, which forwards samples and duplicates nothing, so S1 lying above it at large L is not a violation. The co-located baseline is the marker at $L = 1$, and it is the most efficient deployment anywhere in the figure.

a fifth of it. The best distributed point of the whole sweep is S2 at $L = 4$, at 186.4 Mbit J⁻¹, which is 79 % of the baseline; S3 at $L = 4$ reaches 175.8 and S1 at $L = 16$ reaches 156.0 Mbit J⁻¹.

The ceiling of Remark 1 holds throughout and becomes informative only where it matters. S2 and S3 stay below $1/(\Pi_{FH}L)$ at every point, but the bound is loose where the fronthaul is not yet dominant and tightens as it becomes so: S2 sits at 9 % of the ceiling at $L = 2$ and at 56 % at $L = 128$.

V. CONCLUSION

TODO future work:

- include channel estimation error in rate calculation
- include channel estimation power consumption in power model, and fronthaul consumption of sending pilots to cpu
- include pilot contamination in rate calculation

- no AP selection, all APs are active, investigate the impact on energy efficiency
- make some parameters more realistic e.g. fronthaul power consumption, synchronization power consumption, etc.
- investigate the impact of hardware impairments on system performance
- accurately characterize all parameters based on the size of the BS/APs.
- note that fronthaul consumption is dependent on how it is implemented, this should be further explored. for instance there is the work on serial/daisy chained fronthaul, reference it.
- clearly mention that results depend on current assumptions so they may change, however code is open source and modular so can be used to assess different configurations
- no energy saving techniques are considered or even AP selection, every AP serves every UE currently, investigate the impact of energy saving techniques on energy efficiency

REFERENCES

- [1] E. Peschiera, S. Yun, Y. Lee, L. V. der Perre, and F. Rottenberg, "A parametric power model of upper mid-band (fr3) base stations for 6g," in *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 21 476–21 480.
- [2] 3GPP, "Study on network energy savings for NR (release 18)," 3rd Generation Partnership Project (3GPP), Technical Report TR 38.864, Mar. 2023, version 18.1.0.
- [3] —, "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project (3GPP), Technical Report TR 38.901, version 16.1.0, Dec. 2020. [Online]. Available: <https://www.3gpp.org/DynaReport/38901.htm>
- [4] J. Hoydis, S. Cammerer, F. Ait Aoudia, A. Vem, N. Binder, G. Marcus, and A. Keller, "Sionna: An Open-Source Library for Next-Generation Physical Layer Research," *arXiv preprint*, no. arXiv:2203.11854, 2022. [Online]. Available: <https://arxiv.org/abs/2203.11854>
- [5] H. Q. Ngo, L.-N. Tran, T. Q. Duong, M. Matthaiou, and E. G. Larsson, "On the Total Energy Efficiency of Cell-Free Massive MIMO," *IEEE Transactions on Green Communications and Networking*, vol. 2, no. 1, pp. 25–39, 2018.